

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		Lock and key (1)	1			1		
		(ii)		{Sequence / order / chain} of amino acids (in protein) / folding of {amino acid chain / protein} (1)	1			1		
	(b)	(i)		- Less (intensity of) light passing through (beaker / solution) / light {has been blocked / cannot pass through} / harder for light to pass through (1) - Iodine has reacted with starch / solution in beaker has turned {blue / black / opaque / darker} (1)		1	1	2		1
		(ii)		92 = 3 marks If incorrect award 2 marks Correct readings from graph and substitution into formula (620-160)/5 or 460/5 If incorrect award 1 mark Correct readings from graph 620 and 160 460		3		3	3	
		(iii)		- carbohydrase (has been added / breaks down) (1) (so) starch (is broken down) to glucose (1) Accept maltose {More / easier for} light passing through (solution beaker) / solution becomes {lighter / less opaque} / solution turns from {blue / black} to {orange / brown} / {blue / black} colour {starts to clear / disappears} (1)		2	1	3		3

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		(iv)		Any two (x1) from: • {volume / 100cm³} of starch (1) • {volume / 20cm³} of carbohydrase (1) • {number / amount} of drops of iodine / volume of iodine (1) • {concentration / (1)%} of starch (1) • {concentration / (10)%} of carbohydrase (1) • concentration of iodine (1) • temperature (1) • pH (1) • distance of lamp from <u>sensor</u> / light intensity <u>of lamp</u> (1)	2			2		2
		(v)		Line on graph remains at same level from 10 minutes (1)			1	1		
				Question 8 total	4	6	3	13	3	6